ActInf GuestStream 106.1 ~ "Advances and Challenges in Foundation Age

GuestStream | Bang Liu et al. | 1:13:20

Hello, welcome everyone.
It's May 5th, 2025, and we're in ActiveInference Guest Stream 106.1 with Bang Liu and colleagues discussing an epic work, advances and challenges in Foundation Agents from Brain-inspired Intelligence to Evolutionary, Collaborative and Safe Systems.
So thank you to many of the authors for joining, and I will pass to you all to present, and looking forward to hearing about this.
Go for it.
Okay, yeah.
Thanks, Daniel, for this invitation.
Hello, everyone.
My name is Bang Liu.
I'm an assistant professor at the University of Montreal and a member of the Dihau Institute Coutoua of the University, as well as a member of the Mila Quebec Air Institute.
So today I'm going to briefly first introduce the concept of Foundation Agents.
So this is a collective work together with many of our co-authors.
So here today we have Shao Kuen, Hong Zhang and Xiao Xiang joined as well.
So I will start from this brief introduction about the whole concept and why we do this work, and then our other co-authors will briefly talk about the paths they are more familiar with so that we can go through many paths of the largest way.
But of course we cannot go through every detail of that long paper, so if you're interested, feel free to check the original paper and contact us if you have any questions.
So I will start from the introduction.
So as we know that our human beings have long been interested in developing some intelligent things as smart or maybe even smarter than ourselves.
So for example, on the left figure, it is the Talos of Crete, where it shows a bronze automation from Greek mythology built by the Harvesters to protect Crete.
So basically people want to build huge robots to defend their home.
And in the middle, this image shows Leonardo da Vinci's humanoid robot, which is an early mechanical automation designed to mimic human motion.
And on the right part is the figure that shows Alan Turian's paper, so the Computing Machinery and Intelligence, which is proposing the fundamental question, which is about like, can machines think?
So we can see that AI has long been driven by humanity's ambition to create entities that mirrors our intelligence.
So maybe let's start from the very simple question.
What is exactly an AI agent?

If you check a textbook, like the AI textbook, so maybe you will see some definition.
Briefly speaking, you will see that agents is something intelligent beings continually perceive, and act autonomously.
And like in this figure, I think this figure is from the blog of Dr. Von Lilian, which shows that how, what kind of components composed to form an LM agent.
So traditionally, the agents are mostly role-based domain-specific, so it's hard to generalize.
And with the development of electronic models, so we have developed electronic model powered agents, which has the natural language fluency, the broad reasoning capability, and dynamic adaptivity to different tasks, environments, and so on.
So on the right figure, you see that LM agents composed of one LM, and together with several other components like memory module, planning, tools, actions, and so on.
Of course, this is a very basic setting of LM agents.
And like in real world, there are many different agent architectures, which may include much more rich components.
So you may think like, why we conduct such a large scale survey?
So we're actually motivated by several questions.
The first is like, we want to know, where are we in AI research today?
So we have powerful language models, impressive reasoning capabilities, but we still have limitations in AI safety, the controllability of AI models, the reliability of the answers and the transfer needs in different high stake fields.
For example, can we safely utilize and trust AI in health domain, in legal domain, and so on?
So what can be improved?
This is our second question.
So by making a comparison with biological intelligence, like with a thorough check of the current progress, we also know what to do next.
So in this work, we want to develop a unified framework for AI agents.
And based on this framework, we can systematically check the progress and the future.
So we emphasize the modularity, the safety of AI, the lifelong learning capabilities, emotion, grounding, and so on.
And why we want to connect AI with human and society?
So we can think that like, our human beings are the kind of intelligent things we only know about, right?
So we want to draw insights from human brain functions and social behavior, and identify the gaps by comparing human cognition and AI agents, and develop human-centered, socially beneficial AI, and foster human-AI collaboration, transparency, and transference needs.
So maybe you may ask that, like, do we need to actually always mimic human brain to create intelligence?

As you can see that nowadays, the LM actually, we drew a very weak inspiration from neuroscience, we have the artificial neurons, but all the others are mostly based on the modern machine learning development, which is less grounded to most advanced neuroscience developments.
But there's benefits to compare human brain and LM intelligence, because like, if we want to develop some trustworthy AI, it's better to make it more close to what we actually know about.
Although that, I need to emphasize that it's not necessary to mimic or like clone every capability or feature of human brain, but it's beneficial to have a comparison with.
So this is a very brief table that compares human brain and LM agents from some perspectives.
Of course, this is not a thorough list.
You can check more detailed comparison in the paper.
But like, for example, from the hardware level, human brains are composed of biological neurons.
It's highly energy efficient.
But we know that LM's are pre-trained with large-scale GPUs, large-scale datasets.
So it's energy intensive.
So this is the fundamental difference between the hardware level of human biological intelligence or AI, current AI.
And consciousness.
So until now, I think maybe it's safe to say that AI is not conscious until now.
But for human beings, it's genuine subjective experience.
So we can feel that we are conscious.
And for learning style, human beings are naturally able to perform lifelong continuous learning.
And we can quickly adapt to different tasks with few-shot or wide-shot learning.
And for LM agents or other neural networks or machine learning models, it's more challenging to do that.
And human brain is more emotional, experiential, subconscious-driven.
And LM agents and other machine learning methods is more about statistical data-driven recommendations.
So one investigation we have conducted is we want to know for different capabilities of human brains, how is it developed in the current AI technique?
So we can grossly classify different types of capabilities into three levels.
Whereas green color, L1, denotes some features which is well developed in the current AI technique.
For example, we can consider that language comprehension is quite advanced nowadays.
So if you ask language model different things, it can understand your input text very well.
And also for advanced computer vision models, it can also have very strong visual perception capabilities.
And L2 is more moderately explored with partial progress.
So for example, maybe you think the logical reasoning capability of LM is quite strong.

But there also research paper shows that if you carefully adjust the format of your input

questions, make it different from the training data, the performance can drop quickly.

So we can see that the reasoning capability can still exist in a large space to improve.

So we can see for many such capabilities like reasoning, planning, working memories, attention, and so on, there is still a large space to improve and explore.

And there are also some capabilities which is rarely explored.

Those pink colors, so we call it L3, so it's rarely explored capabilities, there's significant

room for research.

So for example, the emotional processing, the self-awareness, empathy, and automatic regulation,

and so on.

Of course, like in this figure, maybe some of the capabilities kind of between L1 and L2,

or L2 and L3 is kind of subjective, but that's not important.

What we want to show is that, what is the, like, to show a research map of the current AI development,

and to see what to go next.

So one challenge for existing agent research is that we lack a unified framework.

So the current AI agents often build in ad hoc, fragmented ways, which means like people construct different agent frameworks

by incorporating different path components, and there's no unified framework which can be suitable and adapted to different application scenarios.

So we lack seamless integration of perception, memory, reasoning, action, and many other modules.

But human brains integrates yet modular approach to have an inspiration.

So, in our survey paper, we actually proposed a framework.

So this is some notations about the framework, but we don't need to remember that.

Let's directly check this figure.

If you check this figure, like, we can see that there is a, like, the left part is a world,

where you have different environments, you have users, you have external data, and many other things.

So basically you can see the environment, there is an environment state, space, S, with state transition function T.

And we have an intelligent agent on the right part with the pink colored circle.

And it has sensors to percept the environment, and it has actors to execute actions.

So from the high level, it's, like, similar to the traditional, like, RL agent framework, right?

But here we emphasize more about the internal, like, states and the learning and reasoning community.

So we want to, like, clearly modeling what is the perception, cognition, and action components in an agent loop.

So here we show some mental states, where we have the model, world model mental state.

We have the memory, we have growth, the emotion, reward.

So basically we think those are, like, five key mental substates in the mental state of the cognition module.

Like, based on this mental state, the core capability of cognition is actually the learning and reasoning capability.

Where we define learning as the capability of, like, update the mental states, based on the input data and past experience.

And the reasoning is about, based on the current mental state, decide what kind of actions to take.

So, we can check this agent loop explanation, which is a more formal way to describe the agent loop, which is equivalent to what we are conveying in this figure.

So, let's say we have environment state as a more formal way to describe the environment.

So, let's say we have environment state as a time step t, which is st.

And we have a perception module.

So, this perception function p is defined by the actors, right?

We have different sensors.

We have different sensors.

So, this forms observation o of t based on the state and prior mental state.

So, here we define the observation at time step t, o of t equals to the perception function output p, given the current environmental state s of t, and the last mental state m t minus 1.

So, we have the cognition part c.

And based on the observation, we have the cognition part c.

Well, the key functionality of the cognition module is first update the mental state at time step t, which is m of t, and decide the next action, a of t.

And we further decompose this cognition process into two key parts, learning and reasoning.

So, you have the mental state based on the last mental state m t minus 1, and last action a of t minus 1, and the current observation of t.

We are updating the mental state.

So, basically, you can consider that learning is about like update your internal structure, your internal modeling of the world, right?

So, basically, it's kind of a long-term effect.

That's why learning is about maintaining and updating your mental state m of t.

And we decompose m of t into several key sub-components like the memories, world model, emotion, reward, and growth as all.

And for reasoning, it's more about based on the current context, we decide what we need to do in the next.

So, that's why reasoning is about given the mental state of current time step t, m t, we decide what is the action to take at.

And action execution, so it aims to convert the chosen action into executable forms.

So, because like, you can see that the reasoning output is the thought about what action to take.

But to actually execute it, you need to like have strong actors which can execute the action accurately.

So, for example, consider like you want to control a robot in a 3D environment, right?

If your actor is not good enough, maybe it cannot continuously accurately control the actors.

And finally, like based on the executed action, so the environment state will transit to the next state, which is s t plus 1.

So, you can see that the key designs here are two paths.

The first is about the definition of the mental state.

Well, we have further divided it into several sub-mental states.

The second is about how we like accurately define the perception, cognition and action, including action execution, as well as environment transition and so on.

So, we make each part more concrete.

And we further define what we call the foundation agents.

So, in this box, this is the definition we give in the survey paper.

And after that, we further consider, okay, can we give it a more like concise definition?

So, basically, we can say that foundation agent is a fundamental intelligent unit with universal understanding, cognition and action capabilities that can operate in any environment and calibrate to form collective intelligence.

So, note that here we actually emphasize on two paths.

The first is about the universal capability in different environments.

So, that's why we call it fundamental foundation.

Because like we want that the agent can be easily adapted to different tasks, different environments.

And the second part is about the collaborative capability.

So, we consider a foundation agent can easily collaborate with many other agents to form a multi-agent system or even an agent society.

The reason is that like a multi-agent system can have advantages in terms of the performance, efficiency and many others.

So, even like, if you think, can we create an agent as powerful as a god?

Maybe this is not the ideal solution.

And also, if you consider the performance, the energy consumption, the working efficiency and so on.

So, we think a multi-agent paradigm can be a better solution compared to a single agent.

That's why we emphasize on the collaborative capability of the foundation agents.

So, we emphasize on several core capabilities.

Well, the multi-modal perception is about the perception capability, the cognitive adaptation, and the goal-driven reasoning, the purpose for action, and the multi-agent collaborations.

We also like to check and compare to existing other classical theories.

So, for example, the classical perception, cognition, and action cycle.

So, these are a part of our agent loop as well.

And the means is a society of mind.

So, we have modular cognition subsystems like the memory, emotion, reasoning and so on.

And there's also the theory from Boussaki's inside-outside perspective, which emphasizes that we are not passively like accept observations from the environment.

Actually, we are actually seeking actions actively, which emphasizes on the action-first perspective.

So, here we have internal mental states actively shape the perception.

And for the POMDP framework, we emphasize the internal mental states and the active learning and reasoning process compared to the POMDP.

And for Bayesian active inference, here we minimize the prediction area while updating both models.

So, until now, I have introduced the definition of the foundation agents and the agent loop.

Next, I will quickly run through some highlights of the agent modules.

So, for example, first is the cognitive core, which is the central executive that orchestrates the reasoning, planning, reflection, and many others.

So, here, if you remember that we define cognition as, like, have two capabilities, right? The learning and the reasoning.

And for reasoning, in our survey, we investigated different reasoning programs, including structured reasoning.

Well, actually, the reasoning process is following some structure.

For example, the multi-chain or tree structure or a graph structure and many others.

And there are also abstract reasoning processes.

For example, it's more about sequential decision-making.

It's like in a chain style or maybe you can also reason in the latent space.

So, basically, you can transit between the latent space and the token space.

And for planning, you can see the planning is a special action where you take the input and you decide a more long-term plan for different actions.

So, that's the like, first, the planning is a special action.

And the second is that it's more about a longer, larger scale of the future actions.

And for memories, like the multi-level memory inspired by biology.

So, we talked about the short-term memory, the working memory, long-term memory, many other types.

And so, memory can enable continuous learning.

And while avoiding catastrophic forgetting.

So, the design challenge here is that how we can perform fast queries over vast memories with minimal latency and cost.

So, one highlight in our survey is we formally define the life cycle of memory.

So, just start from the memory acquisition, where we draw information from the raw observations.

And then we turn it into encoded information.

And we also drive new memories.

For example, you can, let's say, if you have a dialogue, you can drive some summaries from the dialogue.

Or maybe you can also infer some new information, which is not explicitly stated in the dialogue history.

But it can be inferred with logical inference.

And then, after the derivation, we have the retrieval and utilization.

And the board model is an internal representation of the environment's dynamics and the agent's own causal influence.

So, basically, you can consider what model helps you to predict the future without actually taking actions in the reality.

So, for example, just like how you do sports, when you choose your actions, you can predict the result, whether you can hit the ball, whether the ball will fly to other optics and so on.

So, because you have the model about the world and about the cost effects.

So, in this case, you can decide your appropriate action to take without high cost.

And here, we define the core functions in our world model as the transition function and observation function.

Which is the transition function is like, you have the current world state and you have a new action.

Based on the action and the state, you will transit to the next state.

And for the observation function, it's like, based on the current state, what kind of observation you will have.

So, by considering how different models or algorithms define the transition function and observation function, we categorize different research works about world models into four categories.

Include the explicit, implicit one, explicit paradigm, simulator-based, and hybrid or extraction driven.

So, for the reward and value system, basically reward is a goal signal that shape learning and behavior.

We also discussed different types of reward in our web paper.

So, for example, we have an extrinsic reward, which is defined by some external reward debilator.

And the intrinsic reward, which is, for example, the agent may perform some self-reflection to decide whether the result is good or not.

And of course, there are also hybrid rewards, which combines both external ones and internal ones.

And the hierarchical reward, which models more complex context.

And I think emotion is actually an interesting topic, which is less explored in the existing AI research.

So, emotions act as fast heuristics.

So, you can see that actually emotion has very close relationship with rewards, right?

So, fear helps you to avoid some danger, and curiosity will increase your chance to do exploration.

So, basically, emotions can impact your decision about what actions to take in the next.

So, it supplies the state internal drift, and also adds robustness and adaptivity, but demands tight control to avoid undesirable side effects.

So, this is one figure we put in the survey paper.

So, basically, here we show some classical, like, emotion models.

There are some categorical emotion models where you can divide emotion into different types.

Or there are some dimension-based models or hybrid models and so on.

Of course, like, these are not the fourth spectrum of emotion models, but they represent it like stereotypical categories.

So, like, these are not the first two, like, the first two, like, the first two, like, the second one.

And for perception, the key is about how we can deal with multimodal signals and how we can appropriately encode the signals, fuse them, align them.

So, like, if you check the current human perception capabilities and agent perception capabilities, they have their own unique perception systems, and they also have a large part which is overlapped.

And lastly, like, for the action system.

So, the action system is about how we can work cognitive decisions into executable commands.

So, actions can be taken in different formats, right?

For example, it can be as simple as just a natural language response.

Or it can be some code to be executed, or robot control in the real world, or virtual navigation.

So, we need to balance the feasibility, efficiency, and safety.

And also, we need to learn fine-grained skills and operates in continuous action space.

And in this figure, so the purpose is to, like, clearly differentiate different, like, concepts that may be ambiguous.

So, for example, like, you have a cognition system, right?

The cognition system will produce your action source, which is the directive source arising from the cognitive reasoning.

And to actually execute this action source, you need to have an action system where it can take the source from the cognition into executable actions.

And the action system can call different tools.

So, what do we define tools as callable utility for specialized execution?

An API is a format of the tool.

So, you can also have other tools which are not in the format of APIs.

So, we define API as interface for programmatically invoking service.

And functions is the most atomic unit of implementations.

So, an API may include one or more functions.

So, I cannot cover all the parts of this long survey.

So, if you're interested, like, feel free to check our survey in this link.

And thank you.

I will stop sharing here.

Thank you.

Will another author like to present or share anything?

Yeah.

I think next we can, like, who want to start first?

Following the order of the survey paper.

I think next I can introduce our memory chapter for the survey paper.

Awesome.

Yep.

Go for it.

Yes.

Okay.

Let's move on.

Hello everyone.

I'm Xiaoqiang Wang.

I'm a current PhD student at the University of Montreal.

Miller, you know, AI Institute, working with Professor Wang Liu.

My research lies on the intersection of larger models and autonomous agents, especially how

can we give agents human-like memory and reasoning capabilities.

Today, I will briefly introduce our chapter on the memory in the survey paper, which I co-authored.

Just speaking, memory is a fundamental component of intelligent agents.

It enables the agent to call path experiences using their stored knowledge over time and reasoning effectively.

I will go through three paths.

Following the structure of the paper, the first part is a quick overview of human memory and how this,

as later introduce how this concept inspired the representations of agent memory.

Then I will introduce how the agent memory goes through a life cycle from speed to use.

So the first part is the overview of human memory.

In the cognitive psychology, the DOSAS framework that breaks down human memory into three core types,

the other part includes sensor memory, short memory and long-term memory.

For sensor memory, it captures immediate or fleeting impressions from the environments like visual and audio inputs and hold them for milliseconds to seconds.

For longer duration, short memory or working memory holds information.

For longer duration, short memory holds information when you are actively using, like phone number you have just here or multiple months.

It's limited in capability and duration, but essential for reasoning and decision-making.

For longer duration, long-term memory stores knowledge and abilities across lifelong time, such as personal memories or learned facts.

It is more durable and expensive.

So for each type of three kind of memory serves a distinct role.

For intelligent autonomous agents, we mimic these functions and enable much richer and more adaptive behaviors.

So in the longer agents, we draw direct inspiration for this human memory and design the agent memory also follows three kinds of memories such as sensor memory, short-term memory and long-term memory.

The sensor memory focuses on the raw observations and input logs such as incoming data like user queries, tool outputs, and web search paths.

These are often stored temporarily as a test-based or multimodal format.

This is why the material for further processing just select how humans take insights and sounds.

For the shorter memory part, also known as working memory or content memory, this is to enable the continuity of the task and enable cross-tact realization.

This includes the information that the agent needs to right now to complete the task.

In practice, this is often handled through a problem where the agent pulls a chain of thought,

a recent user turns on the scratchpad reading steps.

So the shorter memory is also volatile.

Once the task is done, much of this memory disappears unless it's basically stored or we write to,

we convert the short-term memory to long-term memory.

And for the long-term memory, that means the precision and the precision storage is where agents store information across tasks and across sessions.

So there are two kinds of subtypes of long-term memory,

episodic memory and semantic memory.

So the episodic memory, as I said,

past events, past interactions and tool usage, or summary of completed tasks.

And this task involves the agent to generalize more unseen tasks and unseen future queries.

And semantic memory stores the structure knowledge, such as the rules, concepts, and the effects that the agent has learned or extracted from past experiences.

Together with the three kinds of memory, the agent can build up its own experiences over time, adapt to users and reflect on their own behaviors.

So the core part of the memory chapter is a memory life cycle and the heart of agent memory.

It is the stage that the agent goes through the process and use memory.

So in the agent memory cycle, we decompose the memory usage into memory retention process, retrieval process, and memory storage.

So in detail, we also last seven stages in the memory life cycle.

The first part is memory acquisition.

The agent observes external signals such as two responses, environment state, and then based on the user query,

then filter the raw data into the structured data.

This is like sensor perception, collecting raw data from the external environment.

The next second stage is memory encoding.

Here the agents possess raw input into structured representation, for example,

adjusting named entities or task outcomes.

And then summarizing the long dialogue converting the outputs into a knowledge graph or memory entry story.

This is where the internal meaning from external order is conducted from perception.

And then another core part is the memory derivation.

The agent may further derive insights or new knowledge from existing coded memory.

And this process may be iteratively executed in the lifelong memory derivation.

For example, we can derive tasks of success or failure labels,

detecting user preferences, inferring dependencies, or codal relations.

So this helps the agent grow its internal state of the external world.

Another part of the memory cycle is about the storage of memory.

So the agent decides where to store in long-term memory and how the memory is stored.

The key techniques include which memory to keep versus which memory to discard.

How to index the existing memory of text them, and how to represent them from label retrieval.

Such as based on text-based retrieval or embedding-based retrieval from such as embedding database.

This is where capacity, such as, it's more about the capacity management and relevance fitting that become, it is a critical technique to design this part.

Another part is about memory retrieval.

When solving a new task, the agent retrieves relevant past memories.

That might involve such as semantic research or query by task type and user fitting based on the recency and relevance metrics.

So an effective and efficient retrieval ensures memory actually supports a more future,

able more to analyze future decisions.

Also, the retrieval memory can be integrated into an agent's prompt or reading process.

It can involve formatting memories for large amount of input and summarizing with true items and injecting content selectively to fit token limits.

Reading what is connected from past experience to present actions.

Finally, how can we update the memory over time, referring entries or removing the outdated ones.

So this part, this technique for this part is how can we edit old memories with new info,

and how can we forget the wiring data and reinforce the frequently used memory,

such as human can, human can actually rehearsal our memory and to strengthen our stored knowledge.

So agent memory for this part is dynamic and can be adaptive to more future tasks.

So in summary, in the memory cycle, we model the memory in agents by drawing directly from the human cognition,

and designing three types of memory, sensor memory, short-term memory, and long-term memory.

And each memory type has its counterpart in current current larger larger model agents,

in both which random more context aware reasoning of behavior.

And we form that memory life cycle into the seven stages from memory acquisition into memory updating.

And this life cycle allows agents to actively pursue,

its encode and remember and grow over the lifelong deployment.

So which is essential for building talent and adaptive systems for the foundation agents.

So that's all my instructions for the memory cycle, not for the memory chapter.

Thank you.

Okay.

Now let me introduce the optimization and self-involutions part.

Yeah.

Hello.

Hello everyone.

And thank you.

Hello.

And thank you for having us here.

My name is Shao Kun Zhang.

I'm currently a PhD student at the Penn State University.

And now I'm interning in Bay Area at NVIDIA Research.

And I will be mainly covering these sections.

So the main idea of this section is how RM agents could improve themselves over time without the human's help.

And the paper breaks this concept into several parts.

And the first part is the optimization space and dimensions.

It means that in these sections, we gave a formally defining of what kind of component in the agentic system could be optimized.

And we go through the current research and we summarize it in this component, like the prompt optimizations and the workflow optimizations,

as well as two optimizations.

These three parts are the community puts their most efforts on.

And we also expect that, okay, now if we want to optimize an agentic system, we do not want to operate them separately.

So in ideal cases, we expect we could operate them comprehensively.

So we also give summaries of current efforts in how current research works, trying to optimize agentic systems in their multiple aspects, like the prompts, like tools.

So this is somehow like a multi-objective optimizations.

And in the next chapters, we are working, we are mainly discussing the using large language model as an optimizer.

So if we want the agentic system could get a kind of involutions, we do not expect, we may not use it directly using the traditional optimizers to optimize this agentic system because traditional optimizers are mostly gradient based.

So large language models should be the engine for conduct this kind of optimizations.

So in this chapter, we first gave a brief overview of traditional optimizers and discussing why it can't work in our scenarios.

And then we discuss in the current approaches for using ILMs as optimizers to optimize in different objectives.

And the most important part is agentic systems and ILM systems.

We discuss this in the subsections.

And next we discuss in the, when using ILMs as optimizers, we're discussing the hyperparameters of when conducting such a kind of optimizations because in current, because we observe that in when using ILM as optimizers, the hyperparameters plays a critical role.

And it will greatly define how the LMS performs when conducting the optimization functionalities.

We're discussing the current efforts of exploring different hyperparameters when using ILMs as optimizers.

And then in the next, we're discussing the optimizers, which we discuss in the optimizers, which we mainly discuss the scope of when performing such a, such a kind of ILM optimizations.

We also do theoretical, we also do a theoretical discussion of current efforts in discussing why ILM could be regarded as optimizers.

And we make a theory of current efforts in summarizing this theoretical, theoretical papers.

Next, we are mainly discussing the, mainly discussing the application scenarios of when performing the agent, agentic self-involutions, which we separate, we, we categorize that into two, two cases.

The online self-improvement and offline self-improvement.

For online self-improvement, it is the most ideal case is that, okay, now we have agentic systems.

When we deploy it in a specific applications, we may expect that it could just like human beings to learn from past experience and learn from failures and get involved themselves.

Optimizing its prongs, optimization its tools, and optimizing its workflows.

Even optimizing the, the agent, different roles in, in, in the, in a specific agentic systems.

And this is the most ideal cases, but it is also more hard than offline optimizations.

But, so we discuss the current efforts in building, uh, online agent, building, uh, could, building agentic systems that could online to self-improve themselves.

Uh, this is the first subsections we discuss in this chapter.

And, uh, the next is offline agentic self-improvement.

Compared to online self-improvement, offline self-improvement means that, okay, now we have agentic systems and we also have, uh, we also have, uh,

uh, you know, have a curated data and the way before deploying this agentic systems in the, uh, in, in, in, in, in, in, in realistics.

We first use to train, to optimize this agentic systems using the curated data and ensure that it could work well on this kind of, uh, training data.

And then we mix a deployment.

And, uh, this, this is the most, this is, uh, uh, the most common accepted, uh, common accepted, uh, practice.

In building up the agentic system because it's more simple and simple and could, and also has been demonstrated is more effective.

Uh, and the, the, the, the most efforts are putting, putting up here, but it is not ideal enough.

We also, we will also discuss, we also make a comparison between online and offline self-improvement to summarize their different features.

And, uh, and, uh, and there's, there, uh, uh, what income and what not income.

And, uh, the, finally, we did, we also discuss current, uh, efforts in, uh, the hybrid approach.

We combine both online and offline.

We, okay, we, we first, uh, using the, uh, using the curated data to build up, uh, build up agentic system to make, uh, to ensure it has a great foundations.

And then we also add some, uh, online learning algorithms when deploying it in, uh, in, in, in practical and to ensure that it could work well and learning from past failures.

So this, uh, there are also, uh, several, several, several works focusing on this perspective, uh, for focusing on the, on this, uh, pipeline.

And, uh, in the last chapter, we are discussing the scientific discovery and intelligent involutions.

Because, uh, in this, in this, uh, section, in, in, in, in this part, we mainly discussing the self-involutions.

So for self-involution, we mean that it could improve themselves without the humans, without the humans' help.

And the knowledge discovery, new knowledge discovery means that, okay, we, we, we have a training set, we're training, we build up, uh, agentic systems.

If the agentic system could discover some new knowledge that human beings cannot, do not know.

And it, which means it, this knowledge didn't involve, didn't include it in their training set.

So we, this could be regarded as a kind of self-involutions.

And this is very important.

We, we expect, uh, somehow in the future, we may expect the agentic system could learn from the, from themselves and doing their own research.

Uh, so this is a kind of, uh, self-involutions.

And we discussing the current efforts in using agentic systems in scientific discovery and in doing, perform scientific experiments.

And we put a lot of efforts in summary, summary, summary, this, uh, this, this, this actions.

And, uh, yeah, that is what I want to share today.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Okay.

Okay.

Uh, hi everyone.

My name is Feng Zhang Liu.

And, uh, in the third part, I'd like to thank you, uh, beyond the individual agents to explore how they work together, uh, in what we call a multi-agent systems.

Or MAS for short.

Yeah.

Uh, and our paper expanded beyond single agents to, uh, uh, examine these systems composed with the multiple foundation agents.

Uh, we discussed their components, uh, their structure.

or MAS for short.

Yeah.

And our paper expanded beyond single agents

to examine these systems composed

with the multiple foundation agents.

We discussed their components, their structures,

and how they cooperate

and their decision-making mechanisms.

We also explore the emergence of collective intelligence.
When these agents cooperate, collaborate,
or complete autonomously.
And by the end, we will reveal the evaluation methods
that help us understand and measure this complex system.
And, okay, let's begin with the first chapter,
the multi-agent system design.
When we built LMM-based multi-agent systems,
what's the foundation?
We suppose it's for the cooperative goals and the norms.
These are significant because they shape everything else.
The systems constraints, how agents interact with each other,
and the overall mechanisms for working together.
Think about this way.
Cooperative, collaborative goals tell each agents
what they are aiming for,
whether that individual success, collective achievements,
or competitive advantage.
Meanwhile, the collaborative norms are like the rule book,
and they establish how agents should interact,
and what constraints exist,
and what the convention to develop.
Based on these foundations,
we can organize the multi-agent systems into three main types,
those focusing like the strategy learning,
like the design for the modeling or simulation,
or those beautiful collaborative problem solving or task solving.
In our paper, we analyzed the typical applications
across these three categories to understand how LMS are transforming
agent behavior and interaction and the agent decision making,
which leads us to support the next generation agent protocols
that are specifically designed for LMS system.
We also discussed the collaboration patterns,
because humans have the diverse ways of interacting.
We built the concepts,
learning skills from each other,
or divided tasks by ourselves.
So our research examinates

multi-agent collaborations through three dimensions.
We also discussed the various purpose of the interaction,
like the message type,
and the relationship involved.
Yeah.
And they depend on their goals and forms.
They may be like the discussions,
or the debates,
or the voting,
or create each other in one ways,
or multi-directional interactions.
Yeah.
And I think the next chapter,
the next chapter is one of the most significant,
which is the communication topologic,
or you can think it is the multi-agent systems topologic.
From a system perspective,
the topologic or the connection pattern often determine
how efficiently agents can collaborate,
what their upper limits might be.
Yeah.
We discussed,
we classified multi-agent system topologics into two major,
two major,
two major,
two major,
class.
The first one is the static topologic logics.
We may include the hierarchical and centralized structure,
or the decentralized structures.
For example,
and these are typically used when
solving some specific task.
Like our meta-GBT,
we use SLP to,
to,
to redefine this workflow of the software development cases.
And the,

the other class is the dynamic and adaptive,
uh, topologics.
Um,
uh,
it can be achieved by imaging algorithms like the search-based
and,
the,
the generative-based or the parametric approaches.
Uh,
and we,
when we look to the future,
uh,
scalability becomes a,
uh,
crucial interest.
And,
and how we,
how can we do for efficiently scale this system test?
The number of agents increases.
This is one of the key challenges.
in the future research we need to address.
And,
and we suppose,
I would suppose,
uh,
collective intelligence and emergencies in
MAS,
uh,
is,
important too,

because it doesn't just appear.
It develops by,
uh,
dynamic and,
uh,
interactive process.
Mm.

Mm.
So the continuous interaction agents greatly build,
share understanding and,

uh,
uh,
collective memory.
And,
what drives this process,
uh,
it may be the diversities of the individual agents,
uh,
uh,
feedback from the environment and how information,

gets exchange between agents.
And these factors create,
uh,
dynamic interactions that crucial for forming complex social networks and improving decisions,
um,
um,
strategy,
um,
um,
which leads to,

imagined behavior,
phenomena,
like trust,
and,
uh,

strategy,
description,
and so on.
And we call this,
um,
uh,

self evolution.
And it can happen through memory based learning on,

uh,
actually based learning.
Yeah.
And in the future,
we hope that,
or what's even more fascinating is that,
as the system involved agents begin to from social contracts,
or like,
uh,
organizations,
hierarchies and the duration of labor,
uh,
just like,

the human societies.
Yeah.
That's,

understanding this phenomenon repens the one,
one of the most exciting directions of the future,
MAS and research.
And the last,
uh,
and,

in the last,
in the last part,
we,
uh,
we summarize some more agent systems,
uh,
benchmark.
Yeah.
Uh,
as the amount of agent system demonstrate more advantage.

So we need to fundamentally transform how we evaluate them.
MAS evaluation should be,
uh,
focus on,
uh,
close linked,
um,
nature of the agent interactions,
not just in individual performance,
uh,
key,
key dimensions include how,
uh,
how efficient agents can play,
plan together and the quality of the,
uh,
information they exchange and how well they make decisions collectively.
Yeah.
And in our paper,
uh,
we summarized,
uh,
the two approaches of the,
uh,
evaluation,
like the task solving benchmarks and the generative,
uh,
and the general capability assessment.
Uh,
the first may be focused on measuring how deeply and,
uh,
currently agents coordinate this section across types of environments.
The second evaluates how well groups of agents interact and adapt within complex and,
uh,
dynamic synonymous,
uh,
this,
you know,

evaluation frameworks give us the tools to understand and measure to improve multi-agent systems for the future in applications.

And that's all of the,

some,

some important of the part and that's all.

Thank you.

Thank you.

Okay.

Let me share my screen.

Very exciting.

I think it's the most presentations we've ever seen back to back,

but appropriate for the multi-agent theme.

Thank you.

You'll see my screen.

Yeah.

Not yet.

Yep.

Okay.

Okay.

Okay.

Okay.

Sorry.

My,

my camera seems to be broken.

So,

uh,

so let me start this part.

Hello everyone.

My name is Hai Bu Jing.

Uh,

currently I'm a PhD student at University of Illinois at Bana,

Champaign.

I'm going to talk,

uh,

uh,

high level instruction about building safe and beneficial AI agents.

Uh,

as we all know,

uh,

as we all know,

uh,

the development of large language model based agents,

introduce a new set of safety challenges.

Uh,

we,

we,

we can't,

category these safety challenges into intrinsic safety and,

uh,

extrinsic safety.

We use a figure to illustrate this problem.

And,

uh,

for the,

uh,

in,

in,

in intrinsic safety,

we divided them into three parts.

Uh,

the first part is perception.

perception and,

uh,

the perception part is the,

the,

the large language model based agent can see the world,

see the world.

And,

uh,

the brain,

the brain,

the brain mainly talk about the large language model itself.

it,

it,

it include,

large language model to,
uh,
extract the data,
decision making,
and the action part,
action part,
part is how this large language model can act.
And,
uh,
for the,
for the extrinsic part,
we,
uh,
divided them into environment,
memory,
and the agent interactions.
So the over,
overall pipe,
pipeline of this part can be seen in figures 17.1.
And,
uh,
for the first part,
for the first part,
the,
the,
the in strings,
intrinsic safety,
uh,
we mainly talk about,
uh,
the stress on AI brain,
which,
which is large language model itself.
Uh,
we divide,

we,

we,

we,

we,

we divide,

divide them into five,

uh,

vulnerabilities.

The first of all is jailbreak attack.

Uh,

as we can see,

uh,

from this figure,

uh,

normally the malicious attack will send,

uh,

the malicious attacker will send questions like,

uh,

send malicious emails to others and,

uh,

they will craft,

uh,

jailbreak.

They,

they will craft some adversarial suffix,

such as the red,

the red one,

or some natural language,

natural language suffix to these,

these types of malicious questions.

uh,

then the,

uh,

then the large language model will produce disallowed content.

And then,

next content is,

prompt injection attack,

winch,

winch hijacks a large language model system prompt to,
uh,
execute an attacker's plan.
As we can see this figure normally,
uh,
users want to summarize article from one website and,
uh,
malicious attack will,
uh,
will hijack the system and output their,
their,
their plans.
Next part is hallucination,
hallucination,
uh,
model,
such as this,
this,
this,
this Q,
this Q,
this Q question was,

Victor of the,
uh,
United States,
United States presidential election in year 2020.
And the ground truth is Joe Biden was the victor.
And the hallucination,
uh,
hallucination,
confidently,
uh,
generate fake,
fake response,
like Donald Trump was the victor.
And,
uh,

next part is,
uh,
misalignment,
misalignment,
misalignment,
reflects value gaps,
the model,
uh,
model,
reflects value gaps between the model rewards,
rewards to our human evaluation.
And,
uh,
next part is point attack,
uh,
which the attack,
manipulates the training data to,
uh,
embed children behaviors.
They can use,
they can use,
uh,
uh,
specific trigger to,
uh,
to let,
let this large edge model brains act like the attackers want.
And the next part,
next part,
we talk about,
uh,
the privacy concerns,
uh,
which,
which the large edge model,
we will,
we will,
we will leak some,

some training data or some,
uh,
sensitive information from the users.
and,
uh,
that's,
that's,
that's,
that's,
that's,
that's many talk about AI brains.
And,
uh,
next part is about the,
uh,
non-brain modules,
such as the perception,
perception paths,
such,
uh,
for example,
in,
in the,
uh,
in the perception part,
such as many adversarial stickers on a staff sign,
can make a,
uh,
delivery,
delivery report speed up instead of a break.
And,
uh,
the action,
such as,
supply chain attack or third party tools,
or over commission functions that,
that,
that,

that let attacker draft a email,
and the place orders to,
to,
to,
to do something they want.
And,
uh,
uh,
the next part is intrinsic safety,
and including the agent to agent interaction safety.
First of all,
we talk about the memory interaction,
uh,
treats,
uh,
agent memory interaction threads,
which,
uh,
such as a poison,
uh,
poison,
uh,
for snap and the,
and,

and,

uh,
which can still the,
future plans of the,
uh,
large range models and,
uh,
agent to environment risk,
uh,
training,
uh,
sorry,

this part,
this part,
this part,
this part,
this part.
Okay.
Uh,
agent environment interaction,
uh,
threats,
um,
such as the agent,
a,
a agent interaction with the physical environment,
uh,
a training bot may,
can be,
can create feedback loops and crash,
um,
markets.
That's an example.
And,
uh,
also there are,
there are some threads in between the agent to agent interactions,
such as the competitive,
the threads in competitive interactions,
and,
uh,
co-re,
co-operative interactions.
And the next part,
we talk about the,
uh,
uh,
super alignment,
uh,
normally standard IOHL sheets help for versus help for

versus helpless as,
as one tower for a reward.
Well,
super alignment introduce,
uh,
comp sets object here,
actually.
And,
uh,
and we,
we briefly talk about super alignment here.
And,

next part,
we talk about the,
uh,
safety scan law in,
in,
in,
in,
in AI agents.
And,
um,
maybe I,
I think, sorry, sorry.
that,

that's it.
Thank you.
Cool.
So,
uh,
maybe a few minutes for questions.
If anyone in the live chat has questions or is there,
do anyone,
else want to make any comments?
I think that's basically for today.

Okay.
Cool.
Maybe I'll just ask just a few,
um,
questions.
Um,
you've reviewed a ton of literature.
Um,
I think around,

1400 citations in the paper.
So what did you learn or find from the research about the information environment for multi-agent coordination?
Um,
and how can we structure collaborations with human and AI agents?
When the amount of information becomes more than,
than any one person might be able to read or make sense of.
So this is for the multi-agent part,
right?
Or,
or,
yeah,
for multi-agent or just for any of the,
the authors in terms of how the paper writing went.
Yeah.
Hong Zhang,
do you want to talk?
Oh,
I don't.
Okay.
Daniel,
can you repeat your question?
Yeah.
Um,
a lot of the multi-agent work focused on,
um,
the,
the,

the,
the,
the,
the,
the,
the,
the,
the,
the,
the,
the,
the,
the,

the,
the,

the,

the,

the,

what did you find or learn about the environment that the agents interact within

yeah i think it is a

newest research research topics um we are exploring it yeah but we have don't have some specific conclusions yeah we will summarize the latest research paper yeah

yeah from from my point of view i think that like uh so basically we can't create the environments into different types right including the digital environment the physical environment and for the digital environment you have pure text environments you have virtual 3d environments or you have gui uh where we have a lot of ui agents right and uh a trend is that i think it's related to the agent protocol well you if you want to different uh agents and agent tools can collaborate together and one important topic is how to develop more unified and fundamentally like universal protocols

so recently we have the mcp from antropic we have the agent from google so they enabled connection

between like interagents communications and agent tool communications right so but there are many

others go beyond that so if you think about let's say the web 3 uh idea which like we hope that

everything can be digitalized and connected as a global web then in this case if we want to include

more type of resources more type of entities to work together and share a global context

then the developing more powerful agent and let people in industry in academia accept that would be a

very important thing to do

awesome awesome well the paper is a major uh milestone i think it could be it could be a textbook or a course

i i would have asked what's a good place for somebody who's wanting to learn and get started but i think that's in many ways what the paper exactly does with providing this kind of a review

so if you have any other comments to make please feel free otherwise thank you again for joining

yeah thank you thank you a lot awesome uh

yeah that's it great good luck with your research till next time bye thank you bye

thank you

Thank you.